

SUBMITTAL

Project

Kerr Stock Equipment

Date

Tuesday, July 18, 1905

Designer

applied@kerrcontrols.ca

Notes

KERR STOCK EQUIPMENT

Kerr Applied

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Prepared By: Kerr Applied 09:03AM

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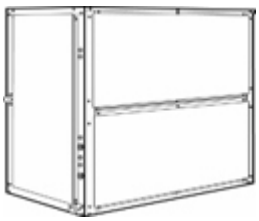
38AUQD07A0M5-0A0A0 + 40RFQA07A2A5-UA0A0

Submittal Cover Sheet
Unit Report
Performance Summary Report
Acoustic Summary
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Feature Sheet

Unit Report For 38AUQD07A0M5-0A0A0 + 40RFQA07A2A5-UA0A0

Project: Kerr Stock Equipment 10/25/2026
Prepared By: Kerr Applied 09:03AM





Outdoor Unit Parameters

Unit Quantity:	**1**	
Unit Model:	**38AUQ**	
Unit Size:	**6 Tons**	
Voltage:	**230-3-60**	V-Ph-Hz
No. of DX Circuits:	**1**	
No. of Stages:	**Dual Stage**	

Indoor Unit Parameters

Unit Model:	**40RFQ**	
Unit Size:	**6 Tons**	
No. of Coils:	**1**	
Voltage:	**230-3-60**	V-Ph-Hz

System Parameter

System Quantity:	**1**
Refrigerant Type:	**PURON**
Compressor Quantity:	**1**
Compressor Type:	**Hermetic**
Std. Capacity Steps:	**67,100**
No. of Outdoor fans:	**2**

Outdoor Unit Dimensions and Weight

Unit Length:	**4' 11.4"**	
Unit Width:	**3' 9.9"**	
Unit Height:	**3' 6.4"**	
Unit Weight:	**393**	lb

Indoor Unit Dimensions and Weight

Unit Length:	**4' 1.0"**	
Unit Width:	**2' 4.2"**	
Unit Height:	**4' 8.1"**	
Unit Weight:	**380**	lb

Warranty Information Outdoor (Note: for US & Canada only)
First Year - Parts Only (Standard)

Warranty Information Indoor (Note: for US & Canada only)
First Year - Parts Only (Standard)

NOTE: Please see Warranty Catalog 808-218 for explanation of policies and ordering methods.

Ordering Information

Part Number	Description	Quantity
Base Unit - Outdoor		
38AUQD07A0A5-0A0A0		1
	Base Unit	
	Al/Cu Condensing Coil	1
	Single Circuit Dual Stage Refrigerant Options	1
	Service Options - None	1
	Electrical Options - None	1
	Packaging Options - Standard	1
	Standard Electrical Mechanical Controls	1
	Refrig Circ/Compressor Staging - Single Circuit/Dual Stage	1
Base Unit - Indoor		
40RFQA07A2A5-UA0A0		1
	Base Unit	
	Std Static, Std Eff Motor / Medium Drive	1

	Cabinet Paint - None	1
	2-Speed Fan Controller (VAF)	1
	Electromechanical Unit Control Board	1

Performance Summary For 38AUQD07A0M5-0A0A0 + 40RFQA07A2A5-UA0A0

Project: Kerr Stock Equipment 10/25/2026
Prepared By: Kerr Applied 09:03AM

System:	**38AUQ007/40RFQ_07**	
No. of Circuits:	**Single Circuit**	
System Quantity:	**1**	
Altitude:	**0.0**	ft
EER @ ARI Conditions:	**11.7**	
IEER:	**15.3**	
COP @ AHRI Conditions:	**3.4**	
Suction Line Loss:	**1.4**	°F

System meets ASHRAE 90.1-2023 min IEER requirement.

This system has a combined part load efficiency (IEER) that meets or exceeds the US DOE 2023 minimum cooling efficiency standards.

Outdoor Unit Parameters

Unit Quantity:	**1**	
PartNumber:	**38AUQD07A0A5-0A0A0**	
Unit Model:	**38AUQ**	
Unit Size:	**6 Tons**	
No. of DX Circuits:	**1**	
Voltage:	**230-3-60**	V-Ph-Hz
Clg Ent Air DB:	**95.0**	°F
Htg Ent Air DB:	**47.0**	°F

Outdoor Electrical Data

Unit Voltage:	**230-3-60**	V-Ph-Hz
Unit MCA:	**25.0**	Amps
Unit MOCP:	**30.0**	Amps
Total Compressor Power of Unit:	**NA**	
Voltage Range Min:	**187**	V
Voltage Range Max:	**253**	V
Compressor RLA:	**17.5**	
Compressor LRA:	**136**	
Compressor Quantity:	**1**	
Fan Motors Qty:	**2**	
Fan Motor FLA:	**1.5**	Amps

Notice: Outdoor unit elect. data is based on 230-3-60
*Single Circuit, Two-Stage Compressor Option

Control Panel SCCR: 5kA RMS at Rated Symmetrical Voltage

Indoor Unit Parameters

PartNumber:	**40RFQA07A2A5-UA0A0**	
Unit Model:	**40RFQ**	
Unit Size:	**6 Tons**	
Coil Rows:	**4**	
No. of Splits:	**1**	
Voltage:	**230-3-60**	V-Ph-Hz
Actual Airflow:	**3000.0**	CFM
Standard Airflow:	**3000.0**	CFM
Total Clg Cap.(Gross):	**73.9**	MBH
Sensible Clg Cap.(Gross):	**63.1**	MBH
Ent Air DB:	**80.0**	°F
Ent Air WB:	**67.0**	°F
Ent Enthalpy:	**31.44**	BTU/lb
Lvg Air DB:	**60.5**	°F
Lvg Air WB:	**59.4**	°F
Lvg Enthalpy:	**25.96**	BTU/lb
Coil Bypass Factor:	**NA**	

Heat Pump System

Htg HP Capacity:	**69.56**	MBH

Htg HP Integrated Capacity: **69.56** MBH
Heating HP Compressor Power: **0.00** kW
Htg Ent Air DB Temp: **70.0** °F
Htg Lvg Air DB: **91.5** °F
COP: (AHRI Conditions): **3.4**

Indoor Supply Fan

Indoor Unit External Static: **0.60** in wg
Economizer Loss: **0.00** in wg
Grille Loss: **0.00** in wg
Plenum Loss: **0.00** in wg
Dehumidification Loss: **0.00** in wg
Acc. Heating Loss: **0.00** in wg
Total Ext Static: **0.60** in wg
Fan Speed: **1338** RPM
Fan Power: **0.86** BHP
Fan Motor Max: **2.40** BHP
Fan Motor FLA: **5.8** Amps
Motor and Medium Static Drive Required.

Indoor Electrical Data

Unit Voltage: **230-3-60** V-Ph-Hz
Unit MCA: **8.0** Amps
Unit MOCP: **15.0** Amps
Notice: Indoor unit elect. data is based on 230-3-60

Acoustics

Sound Power Levels, db re 10E-12 Watts

A-Weighted	Outdoor Unit (dB)	Indoor Unit (dB,Ducted)
Total Level	84.5	86.3
63Hz	65.2	67.0
125Hz	71.7	73.1
250Hz	76.3	76.6
500Hz	80.0	81.0
1000Hz	78.7	80.2
2000Hz	75.7	79.4
4000Hz	69.0	75.2
8000Hz	62.7	NA
Sound Message	Sound for AUQ_07	Sound for RFQ_07

Acoustic Note:

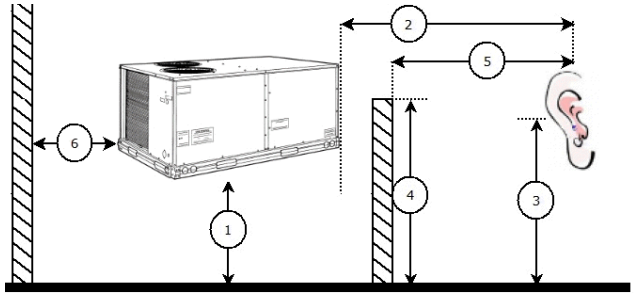
1. 38AXZ/D,38AUZ/D/Q units sound ratings are in accordance with AHRI 270-2008 - Sound Rating of Outdoor Unitary Equipment.
2. 40RL,40RF/40RU units sound ratings are in accordance with AHSRAE 1987 HVAC Systems and Applications handbook.
3. The acoustic center of the unit is located at the geometric center of the unit.

Performance Summary For 38AUQD07A0M5-0A0A0 + 40RFQA07A2A5-UA0A0

Project: Kerr Stock Equipment 10/25/2026
Prepared By: Kerr Applied 09:03AM

Advanced Acoustics Parameters

1. Unit height above ground: **1.0** ft
2. Horizontal distance from unit to receiver: **20.0** ft
3. Receiver height above ground: **5.7** ft
4. Height of obstruction: **0.0** ft
5. Horizontal dist. from obstruction to receiver: **0.0** ft
6. Horizontal dist. from unit to obstruction: **0.0** ft



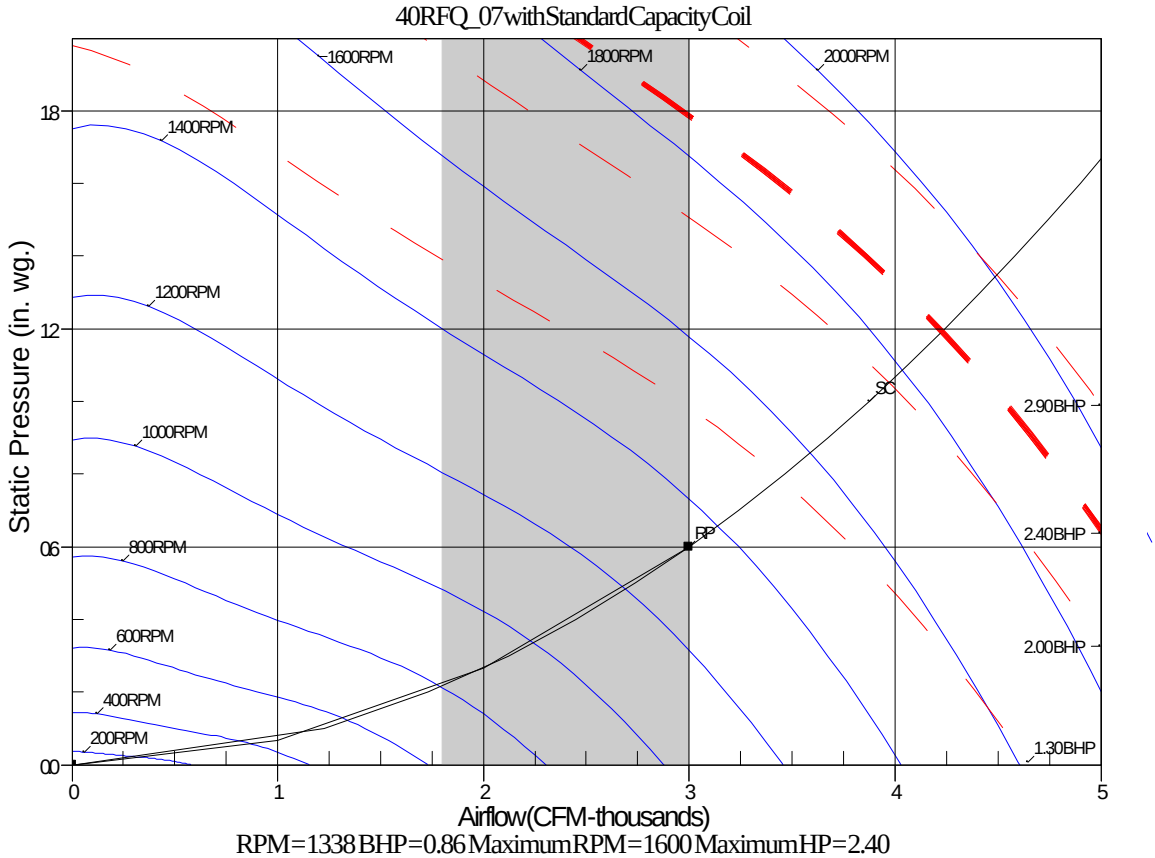
Detailed Acoustics Information

Octave Band Center Frequency, Hz	63	125	250	500	1k	2k	4k	8k	Overall
Sound Power Levels at Unit's Acoustic Center (Lw), dB	91	88	85	83	79	75	68	64	94
A-Wgted Sound Power Levels at Unit's Acoustic Center (LwA), dBA	65	72	76	80	79	76	69	63	85
Sound Press. Levels at Dist. Specified above (Lp), dB	67	63	61	59	54	50	44	39	70
A-Wgted Sound Press. Levels at Dist. Specified above (LpA), dBA	41	47	52	56	54	51	45	38	60

Calculation methods used in this program are patterned after the ASHRAE Guide; other ASHRAE Publications and the AHRI Acoustical Standards. While a very significant effort has been made to insure the technical accuracy of this program, it is assumed that the user is knowledgeable in the art of system sound estimation and is aware of the tolerances involved in real world acoustical estimation. This program makes certain assumptions as to the dominant sound sources and sound paths which may not always be appropriate to the real system being estimated. Because of this, no assurances can be offered that this software will always generate an accurate sound prediction from user supplied input data. If in doubt about the estimation of expected sound levels in a space, an Acoustical Engineer or a person with sound prediction expertise should be consulted.

Performance Summary For 38AUQD07A0M5-0A0A0 + 40RFQA07A2A5-UA0A0

Project: Kerr Stock Equipment 10/25/2026
Prepared By: Kerr Applied 09:03AM



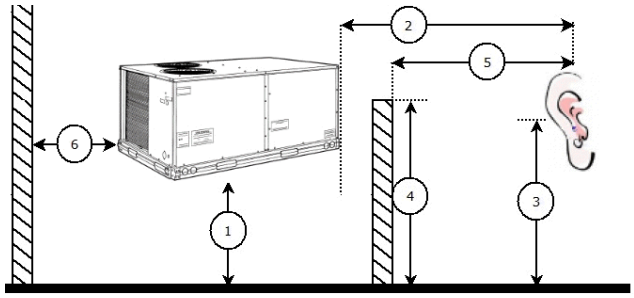
RPM=1338 BHP=0.86 Maximum RPM=1600 Maximum HP=2.40
Shaded Area-Recommended Operating Range
SC-System Curve RP-Rated Point MP-Minimum Point

Acoustic Summary For 38AUQD07A0M5-0A0A0 + 40RFQA07A2A5-UA0A0

Project: Kerr Stock Equipment 10/25/2026
Prepared By: Kerr Applied 09:03AM

Outdoor Unit Parameters:

Tag Name: **38AUQD07A0M5-0A0A0 + 40RFQA07A2A5-UA0A0**
Unit Model: **38AUQ**
Unit Size: **6 Tons**
System Type: **Dx Heat Pump**
Refrigerant Type: **PURON**
Compressor Quantity: **1**
Compressor Type: **Hermetic**



Advanced Accoustics Parameters

1. Unit height above ground: **1.0** ft
2. Horizontal distance from unit to receiver: **20.0** ft
3. Receiver height above ground: **5.7** ft
4. Height of obstruction: **0.0** ft
5. Horizontal distance from obstruction to receiver: **0.0** ft
6. Horizontal distance from unit to obstruction: **0.0** ft

Detailed Acoustics Information

Octave Band Center Frequency, Hz	63	125	250	500	1k	2k	4k	8k	Overall
Sound Power Levels at Unit's Acoustic Center (Lw), dB	91	88	85	83	79	75	68	64	94
A-Wgted Sound Power Levels at Unit's Acoustic Center (LwA), dBA	65	72	76	80	79	76	69	63	85
Sound Press. Levels at Dist. Specified above (Lp), dB	67	63	61	59	54	50	44	39	70
A-Wgted Sound Press. Levels at Dist. Specified above (LpA), dBA	41	47	52	56	54	51	45	38	60

Calculation methods used in this program are patterned after the ASHRAE Guide; other ASHRAE Publications and the AHRI Acoustical Standards. While a very significant effort has been made to insure the technical accuracy of this program, it is assumed that the user is knowledgeable in the art of system sound estimation and is aware of the tolerances involved in real world acoustical estimation. This program makes certain assumptions as to the dominant sound sources and sound paths which may not always be appropriate to the real system being estimated. Because of this, no assurances can be offered that this software will always generate an accurate sound prediction from user supplied input data. If in doubt about the estimation of expected sound levels in a space, an Acoustical Engineer or a person with sound prediction expertise should be consulted.

Acoustic Note:

1. 38AXZ/D,38AUZ/D/Q units sound ratings are in accordance with AHRI 270-2008 - Sound Rating of Outdoor Unitary Equipment.
2. The acoustic center of the unit is located at the geometric center of the unit.
3. All estimated sound power levels, dB re 1 Picowatt should not be guaranteed or certified as being the actual sound power levels.

Acoustic Note:

1. Estimated Sound Power levels - dB re: 1 picowatt
2. Estimated Sound Pressure levels - dB re: 20 micropascal
3. Estimated sound levels given above are assumed to originate at the acoustic center of the unit. The acoustic center of the unit is located at the projection of the condensing unit's geometric center of its base.

4. Sound power levels shown above were determined in accordance with ARI standard 370 for large outdoor refrigeration and air conditioning equipment.

5. Calculation methods used in this program are patterned after the ASHRAE Guide; other ASHRAE Publications and the ARI Acoustical Standards. While a very significant effort has been made to insure the technical accuracy of this program, it is assumed that the user is knowledgeable in the art of system sound estimation and is aware of the tolerances involved in real world acoustical estimation. This program makes certain assumptions as to the dominant sound sources and sound paths which may not always be appropriate to the real system being estimated. Because of this, no assurances can be offered that this software will always generate an accurate sound prediction from user supplied input data. If in doubt about the estimation of expected sound levels in a space, an Acoustical Engineer or a person with sound prediction expertise should be consulted.

Acoustic Summary For 38AUQD07A0M5-0A0A0 + 40RFQA07A2A5-UA0A0

Project: Kerr Stock Equipment 10/25/2026
Prepared By: Kerr Applied 09:03AM

Indoor Unit Parameters:

Tag Name:**38AUQD07A0M5-0A0A0 + 40RFQA07A2A5-UA0A0**

Unit Model:..... **40RFQ**
Unit Size:..... **6 Tons**
System Type:..... **Dx Heat Pump**
External Static Pressure:..... **0.60** in wg
Fan Speed:..... **1338** RPM
Fan BHP:..... **0.86** BHP

Detailed Acoustics Information

Octave Band Center Frequency, Hz	31	63	125	250	500	1k	2k	4k	8k	Total
Sound Power,dB	NA	93	89	85	84	80	78	74	NA	96
A-Weighted Sound Power, dBA	NA	67	73	77	81	80	79	75	NA	86

Acoustic Notes:

1. 40RU units sound ratings are in accordance with AHSRAE 1987 HVAC Systems and Applications handbook.
2. The acoustic center of the units is located at the geometric center of the unit.
3. All estimated sound power levels, dB re 1 Picowatt should not be guaranteed or certified as being the actual sound power levels.

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UNIT	ELECTRICAL CHARACTERISTICS	STD. UNIT WT.		CORNER A		CORNER B		CORNER C		CORNER D		CENTER OF GRAVITY			UNIT HEIGHT
		LBS.	KG.	LBS.	KG.	LBS.	KG.	LBS.	KG.	LBS.	KG.	X	Y	Z	H
38AUZ-07 (RTPF)	208/230-3-60,460-3-60,575-3-60	389	176	141	64	96	44	62	28	91	41	18 [457.2]	24 [609.6]	21 [533.4]	42-3/8 [1076.0]
38AUZ(A,B)08 (RTPF)	208/230-3-60,460-3-60,575-3-60	391	177	142	64	96	44	62	28	91	41	18 [457.2]	24 [609.6]	21 [533.4]	42-3/8 [1076.0]
38AUZ(D,E)08 (RTPF)	208/230-3-60,460-3-60,575-3-60	430	195	142	64	96	44	76	34	111	50	18 [457.2]	24 [609.6]	21 [533.4]	42-3/8 [1076.0]
38AUZ-12 (RTPF)	208/230-3-60,460-3-60,575-3-60	490	222	177	80	120	54	78	35	114	52	18 [457.2]	24 [609.6]	24 [609.6]	50-3/8 [1279.2]
38AUZ-14 (RTPF)	208/230-3-60,460-3-60,575-3-60	598	271	195	88	142	64	110	50	151	68	20 [508.0]	25 [635.0]	24 [609.6]	50-3/8 [1279.2]
38AUD-12 (RTPF)	208/230-3-60,460-3-60,575-3-60	516	234	185	84	117	53	83	38	131	59	19 [482.6]	23 [584.2]	24 [609.6]	50-3/8 [1279.2]
38AUD-14 (RTPF)	208/230-3-60,460-3-60,575-3-60	654	297	214	97	155	70	120	54	165	75	20 [508.0]	25 [635.0]	24 [609.6]	50-3/8 [1279.2]
38AUQ-07	208/230-3-60,460-3-60,575-3-60	444	201	134	61	97	44	90	41	123	56	22 [558.8]	25 [635.0]	13 [330.2]	42-3/8 [1076.0]
38AUQ(A,B)08	208/230-3-60,460-3-60,575-3-60	483	219	162	74	110	50	85	39	125	57	20 [508.0]	24 [609.6]	21 [533.4]	42-3/8 [1076.0]
38AUQ(D,E)08	208/230-3-60,460-3-60,575-3-60	523	237	174	79	118	54	96	44	135	61	21 [533.4]	24 [609.6]	23 [584.2]	50-3/8 [1279.2]
38AUQ-12	208/230-3-60,460-3-60,575-3-60	575	261	186	84	126	57	106	48	157	71	21 [533.4]	24 [609.6]	23 [584.2]	50-3/8 [1279.2]

NOTES:

1. MINIMUM CLEARANCE (LOCAL CODES OR JURISDICTION MAY PREVAIL):
   A. BOTTOM TO COMBUSTIBLE SURFACES: 0 INCHES.
   B. OUTDOOR COIL, FOR PROPER AIR FLOW: 36 INCHES ONE SIDE, 12 INCHES THE OTHER. THE SIDE GETTING THE GREATER CLEARANCE IS OPTIONAL. STANDARD CLEARANCES ON REMAINING TWO SIDES.
   C. OVERHEAD: 60 INCHES, TO ASSURE PROPER OUTDOOR FAN OPERATION.
   D. BETWEEN UNITS: CONTROL BOX SIDE, 42 INCHES PER NEC.
   E. BETWEEN UNIT AND UNGROUNDED SURFACES: CONTROL BOX SIDE, 36 INCHES PER NEC.
   F. BETWEEN UNIT AND BLOCK OR CONCRETE WALLS AND OTHER GROUNDED SURFACES: CONTROL BOX SIDE, 42 INCHES PER NEC.
2. WITH EXCEPTION OF THE CLEARANCE FOR THE OUTDOOR COIL AS STATED IN NOTE 1B, A REMOVABLE FENCE OR BARRICADE REQUIRES NO CLEARANCE.
3. UNITS MAY BE INSTALLED ON COMBUSTIBLE FLOORS MADE FROM WOOD OR CLASS A, B OR C ROOF COVERING MATERIAL.

ITC CLASSIFICATION	SHEET	DATE	SUPERCEDES	CONDENSING UNIT		REV
U.S. ECCN:NSR	1 OF 2	06/29/21	05/08/19	38AUZ/ 38AUD/ 38AUQ-07/08/12/14	38AU500349	J

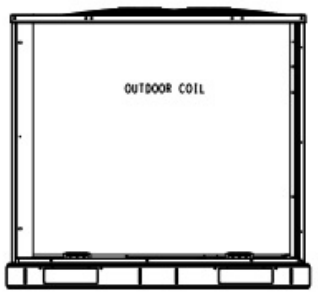
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CG CENTER OF GRAVITY

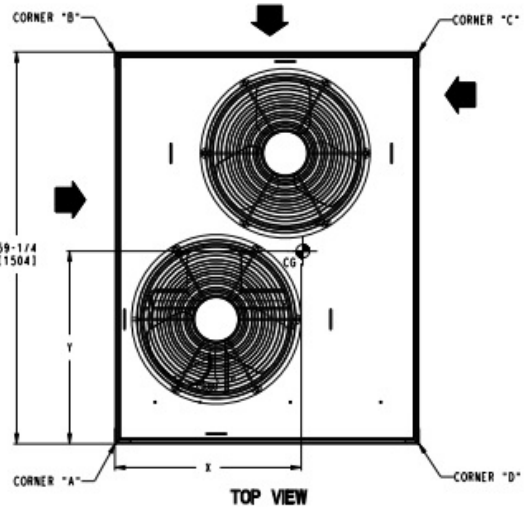
DIRECTION OF AIR FLOW

DIMENSIONS IN [ ] ARE IN MM

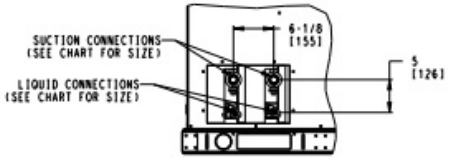
SERVICE VALVE CONNECTIONS			QTY
UNIT	SUCTION	LIQUID	
38AUZ07	1-1/8 [28.6]	3/8 [9.5]	1
38AUZ08	1-1/8 [28.6]	1/2 [12.7]	1
38AUZ12	1-3/8 [34.9]	1/2 [12.7]	1
38AUZ14	1-3/8 [34.9]	5/8 [15.9]	1
38AUD12	1-1/8 [28.6]	3/8 [9.5]	2
38AUD14	1-3/8 [34.9]	1/2 [12.7]	2
38AUQ07	1-1/8 [28.6]	3/8 [9.5]	1
38AUQ08	1-1/8 [28.6]	1/2 [12.7]	1
38AUQ12	1-3/8 [34.9]	1/2 [12.7]	1



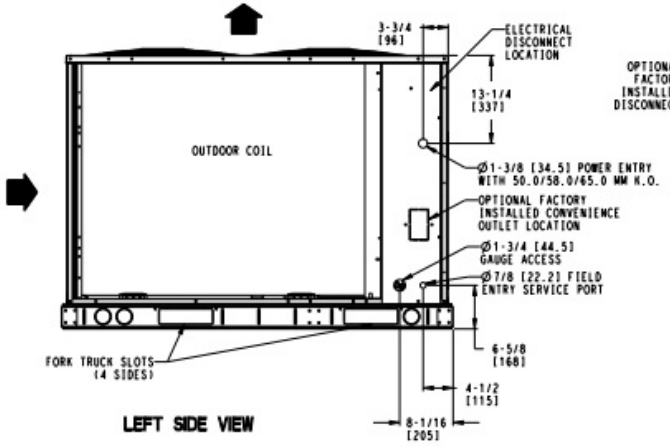
REAR VIEW



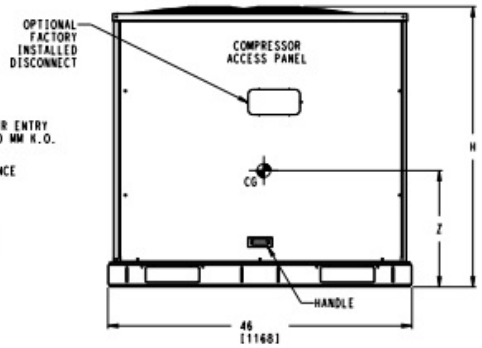
TOP VIEW



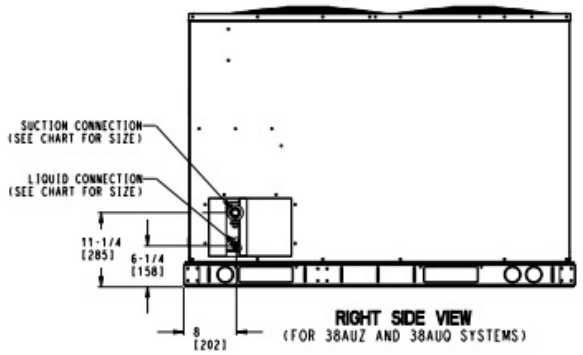
RIGHT SIDE VIEW
(FOR 38AUD SYSTEMS)



LEFT SIDE VIEW

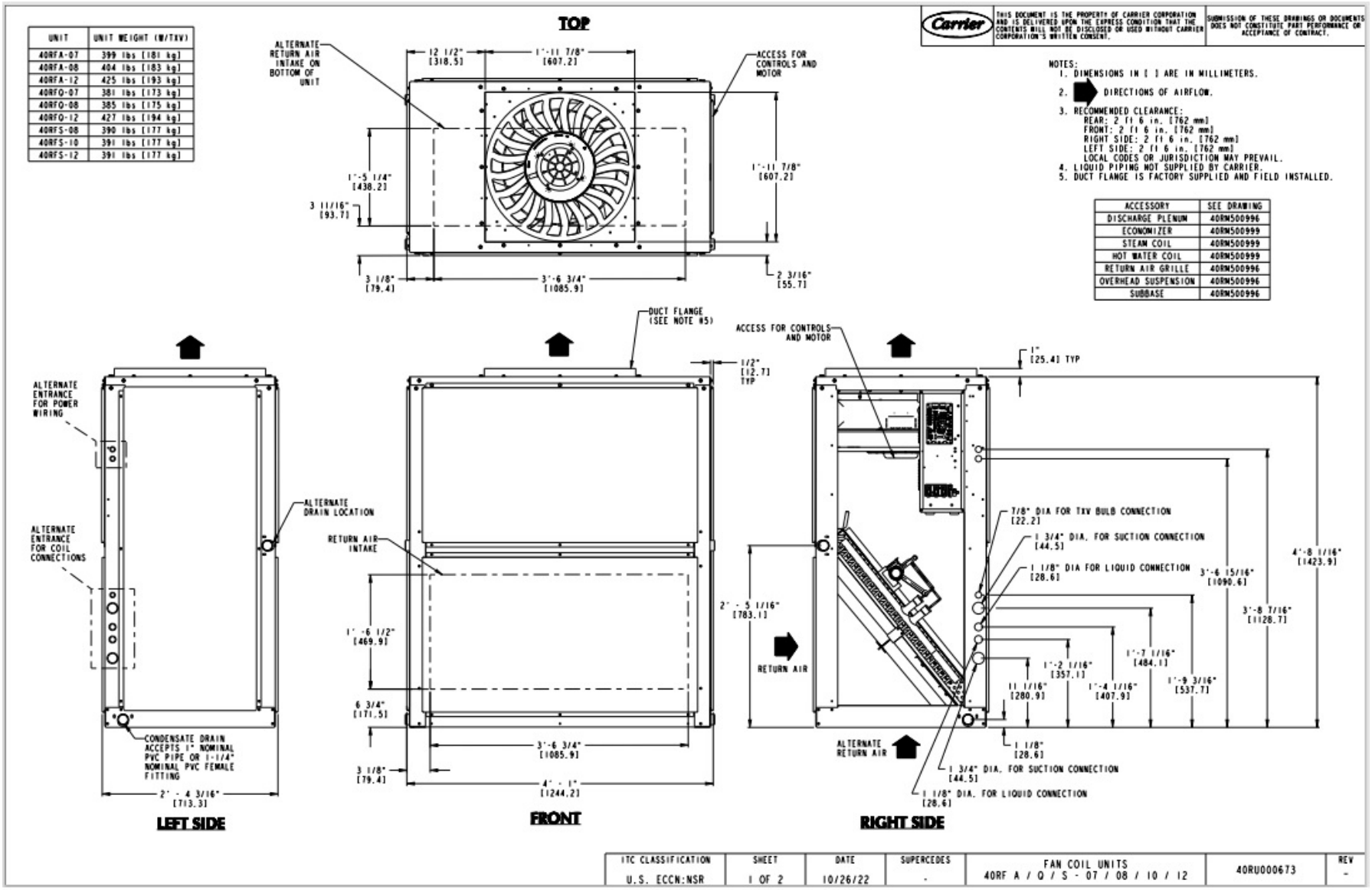


FRONT VIEW



RIGHT SIDE VIEW
(FOR 38AUZ AND 38AUQ SYSTEMS)

ITC CLASSIFICATION	SHEET	DATE	SUPERCEDES	CONDENSING UNIT		REV
U.S. ECCN:NSR	2 OF 2	06/29/21	05/08/19	38AUZ/ 38AUD/ 38AUQ-07/08/12/14	38AU500349	J



UNIT	UNIT WEIGHT (W/TXV)
40RFA-07	399 lbs [181 kg]
40RFA-08	404 lbs [183 kg]
40RFA-12	425 lbs [193 kg]
40RFQ-07	381 lbs [173 kg]
40RFQ-08	385 lbs [175 kg]
40RFQ-12	427 lbs [194 kg]
40RFS-08	390 lbs [177 kg]
40RFS-10	391 lbs [177 kg]
40RFS-12	391 lbs [177 kg]

ACCESSORY	SEE DRAWING
DISCHARGE PLENUM	40RM500996
ECONOMIZER	40RM500999
STEAM COIL	40RM500999
HOT WATER COIL	40RM500999
RETURN AIR GRILLE	40RM500996
OVERHEAD SUSPENSION	40RM500996
SUBBASE	40RM500996

ITC CLASSIFICATION	SHEET	DATE	SUPERCEDES		40RU000673	REV
U.S. ECCN:NSR	1 OF 2	10/26/22	-	FAN COIL UNITS 40RF A / Q / S - 07 / 08 / 10 / 12		-

GUIDE SPECIFICATIONS – 38AUQD07A0A5-0A0A0

Split System Heat Pump Outdoor Unit with PURON® Refrigerant
HVAC Guide Specifications — Section 15678

Size: 07

Part 1: General

SYSTEM DESCRIPTION

1.01. Outdoor-mounted, electrically controlled, air-cooled split system heat pump suitable for on-the-ground or rooftop installation. Unit shall consist of a scroll air-conditioning compressor assembly, an air-cooled coil, propeller-type condenser fans, and a control box. Unit shall discharge supply air upward as shown on contract drawings. Unit shall function as the outdoor component of an air to air electric heat pump system.

1.02. Unit must be designed with a Puron® (R-410A) refrigerant circuit to match Packaged Heat Pump Air Handling Unit(s), 40RUQ for matched systems approved by the manufacturer.

QUALITY ASSURANCE

1.01. Unit shall be rated in accordance with AHRI Standard 340/360.

1.02. Unit construction shall comply with ANSI/ASHRAE 15 safety code latest revision and comply with NEC.

1.03. Unit shall be constructed in accordance with UL 1995 standard and shall carry the UL and UL, Canada label for sizes 07-12 and ETL for 16 and 25 models.

1.04. Unit cabinet shall be capable of withstanding 500-hour salt spray exposure per ASTM B117 (scribed specimen).

1.05. Air-cooled outdoor coils shall be leak tested at 150 psig and pressure tested at 650 psig and qualified to UL burst test at 1980 psig.

1.06. Unit shall be manufactured in a facility registered to ISO 9001:2000 manufacturing quality standard.

DELIVERY, STORAGE AND HANDLING

1.01. Unit shall be shipped as single package only, and shall be stored and handled according to unit manufacturer's recommendations.

WARRANTY (FOR INCLUSION BY SPECIFYING ENGINEER)

Part 2: Products

EQUIPMENT

2.01. General: Factory-assembled, single piece, air-cooled split system heat pump. Contained within the unit enclosure shall be all factory wiring, piping, controls, compressor, holding charge Puron® (R-410A), and special features required prior to field start-up.

2.02. Unit Cabinet:

A. Unit cabinet shall be constructed of galvanized steel, bonderized and coated with a pre-painted baked enamel finish.

B. A heavy-gauge roll-formed perimeter base rail with forklift slots and lifting holes shall be provided to facilitate rigging.

2.03. Fans:

A. Condenser fans shall be direct driven, propeller type, discharging air vertically upward.

B. Fan blades shall be balanced.

C. Condenser fan discharge openings shall be equipped with PVC coated steel wire safety guards.

D. Condenser fan and motor shaft shall be corrosion resistant.

2.04. Compressor:

A. Compressor shall be of the hermetic scroll type.
B. Compressor shall be mounted on rubber grommets.
C. Compressor shall include overload protection.
D. Compressors shall be equipped with a crankcase heater.
E. Compressor shall be equipped with internal high discharge temperature protection.

2.05. Coil:

A. Coil shall be air-cooled and circuited for sub cooling during cooling mode of operation.
B. Coil shall be constructed of aluminum fins (copper fins optional) mechanically bonded to internally grooved seamless copper tubes which are then cleaned, dehydrated, and sealed.

2.06. Refrigeration Components and Operation:

A. Refrigeration circuit components shall include liquid line service valve, suction line service valve, a full charge of compressor oil, and a holding charge of refrigerant.
B. Precision-sized suction line accumulator on each refrigerant circuit shall protect from oil being removed from the scroll compressor rotating orbiter and plate during the activation of the defrost mode and switching back and forth from cooling and heating operations.
C. Unit shall be capable of starting and running up to 125_F (52_C) and down to 35_F (2_C) ambient outdoor temperature.
D. Unit shall operate at +/- 10% from rated voltage

2.07. Controls and Safeties:

A. Minimum control functions shall include:
    1. Control wire terminal blocks.
    2. Compressor lockout on auto-reset safety until reset from thermostat.
    3. Each unit shall utilize the Comfort Alertä Diagnostic Board that provides:
        a. System Pressure Trip fault code indication
        b. Short Cycling fault code indication
        c. Locked Rotor fault code indication
        d. Open Circuit fault code indication
        e. Reverse Phase 3 fault code indication
        f. Welded Contactor fault code indication
        g. Low Voltage fault code indication
        h. Anti-short cycle protection
        i. Phase reversal protection
B. Minimum safety devices which are equipped with automatic reset (after resetting first at thermostat), shall include:
    1. High discharge pressure protection switch.
    2. Loss-of-charge protection switch.

2.08. Electrical Requirements:

A. Unit electrical power shall be single-point connection.
B. Unit control circuit shall contain a 75VA - 24-v transformer for unit control.

2.09. Special Features:

A. Thermostat Controls:
    1. Programmable multi-stage thermostat with 7-day clock, holiday scheduling, large backlit display, remote sensor capability, and Title 24 compliance.
    2. Commercial electronic thermostat with 7-day time clock, auto-changeover, multistage capability, and large LCD temperature display.
    3. Carrier PremierLink™ Controller – Field Installed:
        a. This control will function with CCN and ComfortVIEW™ software. It shall also be compatible with ComfortLink controllers. It shall be ASHRAE 62-99 compliant. It shall accept a CO2 sensor in the conditioned space and be

Demand Control Ventilation (DCV) ready. The communication rate must be 38.4K or faster.

GUIDE SPECIFICATIONS – 40RFQA07A2A5-UA0A0

Commercial Packaged Air-Handling Unit

HVAC Guide Specifications

Size: 07

Part 1: GENERAL

SYSTEM DESCRIPTION

1.01. Indoor, packaged air-handling unit for use in commercial split systems. Unit shall have a multi-position design and shall be capable of horizontal or vertical installation on a floor or in a ceiling, with or without ductwork. (Only vertical units are to be applied without ductwork.)

1.02. Unit with direct-expansion coil shall be used in a refrigerant circuit with a matching air-cooled condensing unit. Unit with chilled water coil shall be used in a chilled water circuit.

QUALITY ASSURANCE

1.01. Coils shall be designed and tested in accordance with ASHRAE 15 Safety Code for Mechanical Refrigeration (U.S.A.), latest edition.

1.02. Unit shall be constructed in accordance with ETL (U.S.A.) and ETL, Canada, standards and shall carry the ETL and ETL, Canada, labels.

1.03. Unit insulation and adhesive shall comply with NFPA-90A (U.S.A.) requirements for flame spread and smoke generation. Insulation shall contain an EPA-registered immobilized antimicrobial agent to effectively resist the growth of bacteria and fungi as proven by tests in accordance with ASTM standards G21 and 22 (U.S.A.).

1.04. Unit shall be manufactured in a facility registered to the ISO 9001:2000 manufacturing quality standard.

1.05. Direct-expansion and chilled water coils shall be burst and leak tested at 435 psi (2999 kPa).

DELIVERY AND STORAGE

1.01. Units shall be stored and handled per manufacturer's recommendations.

Part 2: Products

EQUIPMENT

2.01. Indoor mounted, draw-thru, packaged air-handling unit that can be used in a suspended horizontal configuration or a vertical configuration. Unit shall consist of a direct drive vane axial fan and motor assembly, pre-wired fan motor controller, factory-installed refrigerant metering devices (direct-expansion coil units), cooling coil, 2 in. (51 mm) disposable air filters, and condensate drain pans for vertical or horizontal configurations.

2.02. Base Unit:

A. Cabinet shall be constructed of mill-galvanized steel.

B. Cabinet panels shall be fully insulated with 1/2-in. (12.7-mm) fire-retardant material. Insulation shall contain an EPA-registered immobilized antimicrobial agent to effectively resist the growth of bacteria and fungi as proven by tests in accordance with ASTM standards G21 and 22 (U.S.A.).

C. Unit shall contain non-corroding condensate drain pans for both vertical and horizontal applications. Drain pans shall have connections on right and left sides of unit to facilitate field connection. Drain pans shall have the ability to be sloped toward the right or left side of the unit to prevent standing water from accumulating in pans.

D. Unit shall have factory-supplied 2-in. (51 mm) throwaway-type filters installed upstream from the cooling coil. Filter access shall be from either the right or left side of the unit.

2.03. Coils:

A. DX coil is 4-row and consists of copper tubes with sine-wave aluminum fins bonded to the tubes by mechanical expansion. Suction and liquid line connections or supply and discharge connections shall be made on the same side of the coil.

1. Direct-expansion coils shall feature factory installed thermostatic expansion valves (TXVs) for refrigerant control. The TXVs shall be Puron® R-410A compatible and capable of external adjustment. Direct-expansion heat pump coils shall have a factory-installed bypass line and check valve assembly around the TXVs to allow liquid flow from the coil to the outdoor unit during the heating mode. Coil tubing shall be internally rifled to maximize heat transfer.
2. Chilled water coils shall be rated for an operating pressure of not less than 300 psig (2069 kPag).

2.04. Operating Characteristics:

A. Unit shall be capable of providing ____ cfm (L/s) airflow at an external static pressure of ____ in. wg (kPag).

2.05. Motor:

A. Fan motor of the size and electrical characteristics specified on the equipment schedule shall be factory supplied and installed.

B. Motors rated at 1.3 through 3.7 hp (0.97 through 2.76 kW) shall have internal thermal overload protection. Motors rated at 5 hp (3.73 kW) shall be protected by a circuit breaker.

C. Evaporator-fan motor shall have permanently lubricated, sealed bearings and inherent automatic-reset thermal overload protection or manual reset calibrated circuit breakers. Evaporator motors are designed specifically for Carrier and do not have conventional horsepower (hp) ratings listed on the motor nameplate. Motors are designed and qualified in the "air-over" location downstream of the cooling coil and carry a maximum continuous bhp rating that is the maximum application bhp rating for the motor; no "safety factors" above that rating may be applied.

D. All evaporator-fan motors 5 hp and larger shall meet the minimum efficiency requirements as established by the Energy Policy Act of 1992 (EPACT), effective October 24, 1997.

2.06. Special Features:

A. Alternate Motor and Drive:

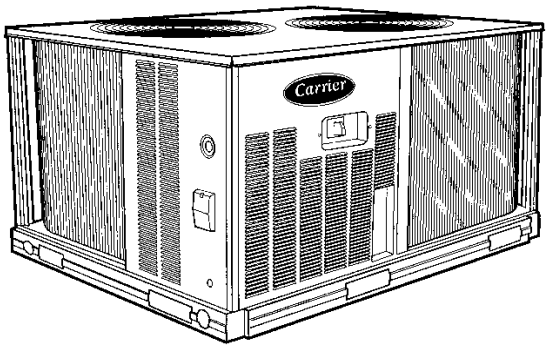
1. An alternate motor and/or medium-static or high-static drive shall be available to meet the airflow and external static pressure requirements specified on the equipment schedule.

B. Thermostat Controls:

1. TEMP System programmable communicating multi-stage thermostat with fan switch, time clock, LCD display, °F/°C capability, and CCN (Carrier Comfort Network™) compatibility.
2. Commercial Electronic Thermostat with 7-day time clock, auto-changeover, multi-stage capability, and large LCD temperature display.

COMMERCIAL SPLIT SYSTEMS
38AUQ 6 to 20 Ton
AIR-COOLED HEAT PUMP CONDENSING UNITS with PURON® REFRIGERANT

These light commercial air-cooled heat pump condensing units are easily connected by refrigerant lines and low voltage control wiring to matching Carrier packaged air-handling units or other suitable evaporator units. They are ideal for new construction or renovation applications where quality and performance are required.



Manufactured in facility registered to
ISO 9001:2008 quality standard

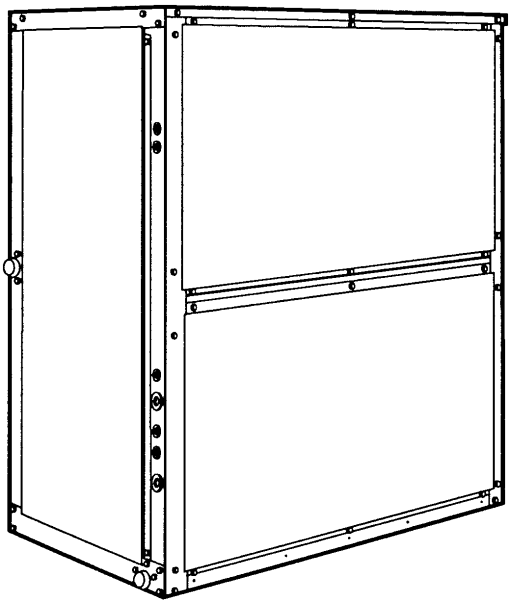
BASE UNIT STANDARD FEATURES:

- Puron® (R-410A) HFC refrigerant partial charge
- 6-10 ton units are single circuit with 2 stage cooling
- 15 & 20 ton units are dual circuit with 3 stage cooling
- All units offer 1 stage heating
- All models utilize round tube plate fin coils.
- Standard one year parts warranty and five year compressor parts warranty. Extended 2-5 year parts and 1-5 labor coverage are available.
- Cooling operating range up to 125°F (52°C) and down to 35°F (2°C) as standard.
- Available Motormaster® I control for cooling operation down to -20°F (-29°C)
- Heating operation up to 60°F (15°C) and down to -10°F (-23°C)
- Fully hermetic scroll compressor with crankcase heater
- New terminal board facilitating simple safety circuit troubleshooting and simplified control box arrangement. 24-volt 75va control system
- High and low pressure switches.
- Comfort Alert™ Diagnostic Board:
    - LED Go-No-Go and fault code
    - Built in time guard anti-short cycle

    - Phase protection
    - Fault code retention logic
    - Low voltage compressor contactor protector
- UL and UL, Canada apply to standard units; 575-volt units UL, Canada only.
- Full perimeter base rails with built-in rigging adapters and fork truck slots
- Pre-painted exterior panels and primer-coated interior panels tested to 500 hours salt spray protection
- Direct drive permanently lubricated condenser fan motors
- All units factory run tested
- Compressors mounted on independent vibration isolators

COMMERCIAL SPLIT SYSTEMS
40RFQ 6 to 10 ton
PACKAGED HEAT PUMP AIR HANDLING UNITS with PURON® REFRIGERANT

The 40RFQ units are completely packaged direct expansion heat pump air-handling units offering the ultimate in installation ease and flexibility. Combined with Carrier 38AUQ Series heat pump condensing units these packaged air handling units offer system capacities from 6 up to 10 tons. These units may be installed in either the vertical or hoizontal configuration with no modifications. Powerful fan systems allow easy adaptation to existing ductwork. Heating capability is provided by electric heaters, hot water coils or steam coils.



ASHRAE 90.1 COMPLIANT

c UL us

AHRI CERTIFIED www.ahridirectory.org Unitary Large AC AHRI Standard 340/360

Gemini

Puron the environmentally sound refrigerant

Manufactured in facility registered to
ISO 9001:2008 quality standard

BASE UNIT STANDARD FEATURES:

- High efficiency Ecoblue Van Axial Fan System
- Unit Control Board (UCB) standard to allow for easy set up
- Puron® (R-410A) HFC refrigerant compatible
- 6 and 10 ton units feature a single evaporator coil
- Designed for vertical or horizontal installation
- Standard sloped drain pans and cleanable insulation
- All models utilize round tube plate fin coils for enhanced cooling performance
- Standard one year parts warranty. Extended 2-5 year parts and 1-5 labor coverage are available.
- High-static design meets a wide range of applications
- Standard factory installed thermostatic expansion valves (TXV) with removable power help to reduce installation time
- Die-formed galvanized steel casings provide durability and structural integrity. Optional paint is available. Optional Staged Air volume (SAV™) system provides two-speed fan control for energy savings
- Ultra LOW LEAK Economizer accessory provides ventilation air and “free” cooling with built in Fault Detection and Diagnostic (FDD) capabilities.
- Hot water coil, steam coil, and electric heat accessories are available.